Ramnatthan Alagappan

Assistant Professor
Department of Computer Science
University of Illinois Urbana-Champaign

Curriculum Vitae - April 7, 2026

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Current Appointments

Assistant Professor
University of Illinois Urbana-Champaign — Aug 2022 – Current

Education

Ph.D. in Computer Sciences
University of Wisconsin – Madison — 2019
Advisors: Andrea C. Arpaci-Dusseau and Remzi H. Arpaci-Dusseau
Thesis: Protocol- and Situation-Aware Distributed Storage Systems

M.S. in Computer Sciences
University of Wisconsin – Madison — 2018

B.Tech in Information Technology
Coimbatore Institute of Technology, Anna University, India — 2010

Honors & Awards

Research	Best Paper Award at SOSP (LazyLog)	2024
	NetApp Faculty Fellowship	2024
	NSF CAREER Award	2023
	Best Paper Award at FAST (CAD)	2020
	UW CS Graduate Student Research Award - Best Thesis - Honorable Mention	2019
	Best Paper Award at FAST (PAR/CTRL)	2018
	Best Paper Award at FAST (CCFS)	2017
	Best Paper Nominee at FAST (Redundancy $\not\Rightarrow$ FT)	2017
Teaching	UIUC List of Teachers Ranked as Outstanding (for CS598 - Storage Systems)	Fall 2023
	UIUC List of Teachers Ranked as Excellent (for CS598 - Storage Systems)	Spring 2023
	UIUC List of Teachers Ranked as Excellent (for CS598 - Storage Systems)	Fall 2022
	CS 739 ranked 1st among all courses in student evaluations	2020
	Nominated for SACM CoW Teaching Award for CS 739	2020
Service	Distinguished Reviewer at HotStorage	2021
	Best Shadow PC Reviewer at EuroSys	2019
Grants	Research Gift from TigerBeetle	2025
	NetApp Faculty Fellowship	2023
	NSF Career Award $\$699,655$	2023
	Co-PI IIDAI IBM grant $\$480,000$	2023
	Microsoft Azure Credits Research Award for $\$50,000$	2019
	Facebook Distributed Systems Research Award for $\$50,000$	2019
	CS Alumni Scholarship, University of Wisconsin – Madison	2013

Peer-Reviewed Publications

Post PhD Work:

OSDI '26 **35.** Jiyu Hu, Seokjoo Cho, Landon Johnson, Kiran Hombal, Shreesha G. Bhat, Marcos Aguilera, **Ramnatthan Alagappan**, Aishwarya Ganesan. *MEGALON: Efficient Data Sharing for Partly Coherent CXL Memory.* In Proceedings of the 19th USENIX Symposium on Operating Systems Design and Implementation, 2025.

EuroSys '26 **34.** Kiran Hombal, Henry Zhu, Shreesha Bhat, Neil Kaushikkar, **Ramnatthan Alagappan**, Aishwarya Ganesan. *A Logically Disaggregated Cache for Replicated Storage Systems.* In Proceedings of the European chapter of ACM SIGOPS, Edinburgh, Scotland. April 2026.

TOCS '25 **33.** Xuhao Luo, Shreesha Bhat, Jiyu Hu, **Ramnatthan Alagappan**, Aishwarya Ganesan. *LazyLog: A New Shared Log Abstraction and Design for Modern Low-Latency Applications* . ACM Transactions on Computer Systems, 2025.
Invited

OSDI '25 **32.** Shreesha Bhat, Tony Hong, Xuhao Luo, Jiyu Hu, Aishwarya Ganesan, **Ramnatthan Alagappan**. *Low End-to-End Latency atop a Speculative Shared Log with Fix-Ante Ordering.* In Proceedings of the 19th USENIX Symposium on Operating Systems Design and Implementation, 2025. Acceptance rate: 53/327 = 16.2%

SOSP '24 **31.** Xuhao Luo, Shreesha Bhat, Jiyu Hu, **Ramnatthan Alagappan**, Aishwarya Ganesan. *LazyLog: A New Shared Log Abstraction for Low-Latency Applications.* In Proceedings of the 30th ACM Symposium on Operating Systems Principles, 2024. Acceptance rate: 43/245 = 17.6%
Best Paper Award
Invited to Transactions on Computer Systems
Ideas influenced a popular streaming platform (Link)

;login: **30.** Xudong Sun, Wenqing Luo, Tyler Gu, Aishwarya Ganesan, **Ramnatthan Alagappan**, Michael Gasch, Lalith Suresh, and Tianyin Xu. *Sieve: Chaos Testing for Kubernetes Controllers.* ;login: The USENIX Magazine, November 2024.
Invited

FAST '24 **29.** Yi Xu*, Henry Zhu*, Prashant Pandey, Alex Conway, Rob Johnson, Aishwarya Ganesan, **Ramnatthan Alagappan**. *IONIA: Efficient Replication for Disk-based KV Stores.* * = equal contribution. (To Appear) In Proceedings of the 22nd USENIX Conference on File and Storage Technologies, 2024. Acceptance rate: 22/123 = 17.8%

EuroSys '24 **28.** Xuhao Luo, **Ramnatthan Alagappan**, Aishwarya Ganesan. *SplitFT: Fault Tolerance for Disaggregated Datacenters via Remote Memory Logging.* In Proceedings of the European chapter of ACM SIGOPS, Athens, Greece. April 2024. Acceptance rate: 71/484 = 14.7%

CACM **27.** **Ramnatthan Alagappan**, Peter Alvaro. *Crash Consistency.* Communications of the ACM - Vol. 66 No. 1, January 2023.
Invited

TOS '22 **26.** Aishwarya Ganesan, **Ramnatthan Alagappan**, Anthony Rebello, Andrea C. Arpaci-Dusseau, Remzi H. Arpaci-Dusseau. *Exploiting Nil-External Interfaces for Fast Replicated Storage.* ACM Transactions on Storage (TOS), May 2022.
Fast-tracked

OSDI '22 **25.** Xudong Sun, Wenqing Luo, Tyler Gu, Aishwarya Ganesan, **Ramnatthan Alagappan**, Michael Gasch, Lalith Suresh, and Tianyin Xu. *Automatic Reliability Testing For Cluster Management Controllers.* In Proceedings of the 16th USENIX Symposium on Operating Systems Design and Implementation, 2022. Acceptance rate: 49/251 = 19.5%

SOSP '21 **24.** Aishwarya Ganesan, **Ramnatthan Alagappan**, Andrea C. Arpaci-Dusseau, Remzi H. Arpaci-Dusseau. *Exploiting Nil-Externality for Fast Replicated Storage.* In Proceedings of the 28th ACM Symposium on Operating Systems Principles, 2021. Acceptance rate: 54/348 = 15.5%
Invited to Transactions on Storage

NVMW '21 **23.** Kan Wu, Zhihan Guo, Guanzhou Hu, Kaiwei Tu, **Ramnatthan Alagappan**, Rathijit Sen, Kwanghyun Park, Andrea C. Arpaci-Dusseau, Remzi H. Arpaci-Dusseau. *The Storage Hierarchy is Not a Hierarchy: Optimizing Caching on Modern Storage Devices with Orthus* Non-volatilve Memory Workshop, 2021.

TOS '21 **22.** Anthony Rebello, Yuvraj Patel, **Ramnatthan Alagappan**, Andrea C. Arpaci-Dusseau, Remzi H. Arpaci-Dusseau. *Can Applications Recover from fsync Failures?* ACM Transactions on Storage (TOS), June 2021.
Fast-tracked

TOS '21 **21.** Aishwarya Ganesan, **Ramnatthan Alagappan**, Andrea C. Arpaci-Dusseau, Remzi H. Arpaci-Dusseau. *Strong and Efficient Consistency with Consistency-aware Durability.* ACM Transactions on Storage (TOS), January 2021.
Fast-tracked

HotOS '21 **20.** Xudong Sun, Lalith Suresh, Aishwarya Ganesan, **Ramnatthan Alagappan**, Michael Gasch, Lilia Tang, Tianyin Xu. *Reasoning about Modern Datacenter Infrastructures using Partial Histories* 18h Workshop on Hot Topics in Operating Systems, 2021.

FAST '21 **19.** Kan Wu, Zhihan Guo, Guanzhou Hu, Kaiwei Tu, **Ramnatthan Alagappan**, Rathijit Sen, Kwanghyun Park, Andrea C. Arpaci-Dusseau, Remzi H. Arpaci-Dusseau. *The Storage Hierarchy is Not a Hierarchy: Optimizing Caching on Modern Storage Devices with Orthus.* In Proceedings of the 19th USENIX Conference on File and Storage Technologies, 2021. Acceptance rate: 28/130 = 21.5%

OSDI '20 **18.** Yifan Dai, Yien Xu, Aishwarya Ganesan, **Ramnatthan Alagappan**, Brian Kroth, Andrea C. Arpaci-Dusseau, Remzi H. Arpaci-Dusseau. *From Wisckey to Bourbon: A Learned Index for Log-structured Merge Trees.* In Proceedings of the 14th USENIX Conference on Operating Systems Design and Implementation, 2020. Acceptance rate: 70/398 = 17.6%

HotStor '20 **17.** Konstantinos Kanellis, **Ramnatthan Alagappan**, Shivaram Venkataraman. *Too Many Knobs to Tune? Towards Faster Database Tuning by Pre-selecting Important Knobs.* 12th Workshop on Hot Topics in Storage and File Systems, 2020.

ATC '20 **16.** Anthony Rebello, Yuvraj Patel, **Ramnatthan Alagappan**, Andrea C. Arpaci-Dusseau, Remzi H. Arpaci-Dusseau. *Can Applications Recover from Fsync Failures?* In Proceedings of the 2020 USENIX Annual Technical Conference, 2020. Acceptance rate: 65/348 = 18.7%
Fast-tracked to Transactions on Storage

FAST '20 **15.** Aishwarya Ganesan, **Ramnatthan Alagappan**, Andrea C. Arpaci-Dusseau, Remzi H. Arpaci-Dusseau. *Strong and Efficient Consistency with Consistency-aware Durability.* In Proceedings of the 18th USENIX Conference on File and Storage Technologies, 2020. Acceptance rate: 23/138 = 16.7%
Best Paper Award
Fast-tracked to Transactions on Storage

PhD Work:

TOS '18 **14.** **Ramnatthan Alagappan**, Aishwarya Ganesan, Eric Lee, Aws Albarghouthi, Vijay Chidambaram, Andrea C. Arpaci-Dusseau, Remzi H. Arpaci-Dusseau. *Protocol-Aware Recovery for Consensus-Based Distributed Storage.* ACM Transactions on Storage (TOS), October 2018.
Fast-tracked

OSDI '18 13. **Ramnatthan Alagappan**, Aishwarya Ganesan, Jing Liu, Andrea C. Arpaci-Dusseau, Remzi H. Arpaci-Dusseau. *Fault Tolerance, Fast and Slow: Exploiting Failure Asynchrony in Distributed Systems* . In Proceedings of the 13th USENIX Conference on Operating Systems Design and Implementation, 2018. Acceptance rate: 47/257 = 18.3%

FAST '18 12. **Ramnatthan Alagappan**, Aishwarya Ganesan, Eric Lee, Aws Albarghouthi, Vijay Chidambaram, Andrea C. Arpaci-Dusseau, Remzi H. Arpaci-Dusseau. *Protocol-Aware Recovery for Consensus-Based Storage.* In Proceedings of the 16th USENIX Conference on File and Storage Technologies, 2018. Acceptance rate: 23/140 = 16.4%
Best Paper Award
Fast-tracked to Transactions on Storage
Invited to ATC 19 Best of the Rest
Runs in production at TigerBeetle

EuroSys'17 11. Amir Saman Memaripour, Anirudh Badam, Amar Phanishayee, Yanqi Zhou, **Ramnatthan Alagappan**, Karin Strauss, Steven Swanson. *Atomic In-Place Updates for Non-Volatile Main Memories with KaminoTx.* In Proceedings of the European Conference on Computer Systems, 2017. Acceptance rate: 41/200 = 20.5%

FAST '17 10. Aishwarya Ganesan, **Ramnatthan Alagappan**, Andrea C. Arpaci-Dusseau, Remzi H. Arpaci-Dusseau. *Redundancy Does Not Imply Fault Tolerance: Analysis of Distributed Storage Reactions to Single Errors and Corruptions.* In Proceedings of the 15th USENIX Conference on File and Storage Technologies, 2017. Acceptance rate: 28/118 = 23.7%
Best Paper Nominee
Invited to Usenix ;login:
Fast-tracked to Transactions on Storage

FAST '17 9. Thanumalayan Sankaranarayana Pillai, **Ramnatthan Alagappan**, Lanyue Lu, Vijay Chidambaram, Andrea C. Arpaci-Dusseau, Remzi H. Arpaci-Dusseau. *Application Crash Consistency and Performance with C2FS.* In Proceedings of the 15th USENIX Conference on File and Storage Technologies, 2017. Acceptance rate: 28/118 = 23.7%
Best Paper Award
Fast-tracked to Transactions on Storage
Invited to ATC 18 Best of the Rest

;login: 8. Aishwarya Ganesan, **Ramnatthan Alagappan**, Andrea C. Arpaci-Dusseau, Remzi H. Arpaci-Dusseau. *Redundancy Does Not Imply Fault Tolerance: Analysis of Distributed Storage Reactions to Single Errors and Corruptions.* ;login: The USENIX Magazine, Summer 2017.
Invited

TOS '17 7. Aishwarya Ganesan, **Ramnatthan Alagappan**, Andrea C. Arpaci-Dusseau, Remzi H. Arpaci-Dusseau. *Redundancy Does Not Imply Fault Tolerance: Analysis of Distributed Storage Reactions to File-System Faults.* ACM Transactions on Storage (TOS), September 2017.
Fast-tracked

TOS '17 6. Thanumalayan Sankaranarayana Pillai, **Ramnatthan Alagappan**, Lanyue Lu, Vijay Chidambaram, Andrea C. Arpaci-Dusseau, Remzi H. Arpaci-Dusseau. *Application Crash Consistency and Performance with C2FS.* ACM Transactions on Storage (TOS), September 2017.
Fast-tracked

OSDI '16 5. **Ramnatthan Alagappan**, Aishwarya Ganesan, Yuvraj Patel, Thanumalayan Sankaranarayana Pillai, Andrea C. Arpaci-Dusseau, Remzi H. Arpaci-Dusseau. *Correlated Crash Vulnerabilities.* In Proceedings of the 12th USENIX Conference on Operating Systems Design and Implementation, 2016. Acceptance rate: 47/267 = 17.6%

HotOs '15 4. **Ramnatthan Alagappan**, Vijay Chidambaram, Thanumalayan Sankaranarayana Pillai, Aws Albarghouthi, Andrea C. Arpaci-Dusseau, Remzi H. Arpaci-Dusseau. *Beyond Storage APIs: Provable Semantics for Storage Stacks.* 15th Workshop on Hot Topics in Operating Systems, 2015.

ACMQueue 3. Thanumalayan Sankaranarayana Pillai, Vijay Chidambaram, **Ramnatthan Alagappan**, Samer Al Kiswany, Andrea C. Arpaci-Dusseau, Remzi H. Arpaci-Dusseau. *Crash Consistency: Rethinking the Fundamental Abstractions of the File System.* ACM Queue, July 2015.
Invited

CACM 2. Thanumalayan Sankaranarayana Pillai, Vijay Chidambaram, **Ramnatthan Alagappan**, Samer Al Kiswany, Andrea C. Arpaci-Dusseau, Remzi H. Arpaci-Dusseau. *Crash Consistency.* Communications of the ACM - Vol. 58, No. 10, October 2015.
Invited

OSDI '14 1. Thanumalayan Sankaranarayana Pillai, Vijay Chidambaram, **Ramnatthan Alagappan**, Samer Al Kiswany, Andrea C. Arpaci-Dusseau, Remzi H. Arpaci-Dusseau. *All File Systems Are Not Created Equal: On the Complexity of Crafting Crash-Consistent Applications.* In Proceedings of the 11th USENIX Conference on Operating Systems Design and Implementation, 2014. Acceptance rate: 42/232 = 18.1%
Invited to Communications of the ACM
Invited to ACM Queue

Research Impact

LazyLog. The ideas from our LazyLog paper have been adopted and implemented in Confluent WarpStream (Link), a data-streaming platform.

Corruption-tolerant Replication. The CTRL protocol from my FAST '18 paper has been adopted and implemented in TigerBeetle (Link1, Link2), a financial database, making it resilient to storage corruptions and errors. This work has also influenced systems at Facebook (Link).

ErrFS and ErrBench. ErrFS is a user-level FUSE file system that systematically injects file-system faults. Ideas from ErrFS have been adopted by other popular testing tools. ErrBench is a suite of distributed-storage-system workloads which drives systems to interact with their local storage. Through ErrFS and ErrBench, we have exposed many serious bugs in popular distributed systems such as ZooKeeper, Cassandra, and Kafka.
Link to Artifacts

PACE. PACE is a framework to systematically generate and explore persistent states that can occur in a distributed execution, exposing crash vulnerabilities in distributed storage systems. PACE found 26 serious, real-world bugs in popular systems including ZooKeeper, Redis, etcd, and Kafka. Many bugs found by PACE have been fixed by developers.
Link to Artifacts

ALICE. ALICE is a crash-consistency testing framework that I helped build. ALICE has been adopted by others (including an open-source version). ALICE found several real-world bugs in 12 widely used commercial storage software products, including Google's LevelDB, Git, and SQLite.
Link

Press Articles on Research

The Morning Paper. Protocol-Aware Recovery for Consensus-Based Storage
Link to Article — Feb 2018

ZDNet. Eliminating Storage Failures in the Cloud
Link to Article — Feb 2018

The Morning Paper. Crash Consistency and Performance with CCFS
Link to Article — Mar 2017

The Morning Paper. Redundancy Does Not Imply Fault Tolerance
Link to Article — Mar 2017

DHSR's Blog. Redundancy Does Not Imply Fault Tolerance
Link to Article — Mar 2017

StorageMojo. Redundancy Does Not Imply Fault Tolerance
Link to Article — Mar 2017

The Morning Paper. All File Systems are Not Created Equal
Link to Article — Feb 2016

Teaching

Instructor, *UIUC*
CS 598 - Cloud Storage Systems — Fall '23
UIUC List of Teachers Ranked as Outstanding

Instructor, *UIUC*
CS 598 - Cloud Storage Systems — Spring '23
UIUC List of Teachers Ranked as Excellent

Instructor, *UIUC*
CS 598 - Cloud Storage Systems — Fall '22
UIUC List of Teachers Ranked as Excellent

Student Advising

Henry Zhu, PhD student, *Started Fall 2022*
Xuhao Luo, PhD student, *Started Fall 2022*
Shreesha Bhat, PhD student, *Started Fall 2023*
Jiyu Hu, PhD student, *Started Fall 2023*
Kiran Hombal, PhD student, *Started Fall 2023*
Emaan Attique, PhD student, *Started Spring 2025*
Landon Johnson, PhD student, *Started Fall 2025*
Michael Noguera, PhD student, *Started Fall 2025*

Wenqing Luo, MS student (graduated)
Cloud-Native Recoverability
Chaitanya Bhandari, MS student (graduated)
Intelligent Block Storage Services
Ramya Bygari, MS student (graduated)
Better Crash Recovery for Replicated Systems

Service

NSDI '27 Program Committee — 2027
ATC '27 Program Committee — 2027
EuroSys '27 Program Committee — 2027

FAST ’27 Program Committee	2027
ATC ’26 Program Committee	2026
OSDI ’26 Program Committee	2026
EuroSys ’26 Program Committee	2026
FAST ’26 Program Committee	2026
SOSP ’25 Program Committee	2025
FAST ’25 Program Committee	2025
SYSTOR ’25 Program Committee	2025
HotStorage ’25 Program Committee	2025
USENIX ATC ’25 Program Committee	2025
PaPoC ’25 Program Committee	2025
SOCC ’25 Program Committee	2025
NSDI ’25 Program Committee	2025
SysDW ’24 Co-Chair	2024
ICDCS ’24 Track Co-Chair	2024
SOSP ’24 Program Committee	2024
USENIX ATC ’24 ERC Co-Chair	2024
USENIX ATC ’24 Program Committee	2024
EuroSys ’24 Program Committee	2024
HotStorage ’24 Program Committee	2024
SYSTOR ’24 Program Committee	2024
Performance ’23 Program Committee	2023
SYSTOR ’23 Program Committee	2023
NVMW ’23 Program Committee	2023
SOCC ’23 Program Committee	2023
HotStorage ’23 Program Committee	2023
OSDI ’23 Program Committee	2023
FAST ’23 Poster/WiP Co-chair	2023
SRC PACT ’22 Program Committee	2022
SOCC ’22 Program Committee	2022
HotStorage ’22 Program Committee	2022
SOSP ’21 Ask-Me-Anything Co-chair	2021
SOSP ’21 Mentoring	2021
OSDI ’21 Mentoring	2021
EuroDW ’21 Mentoring	2021
Journal of Systems SEB Co-chair	2021
EuroDW ’21 Program Committee	2021
HotStorage ’21 Program Committee (**Distinguished Reviewer**)	2021
Systor ’21 Program Committee	2021
ACM Transactions on Computer Systems, Reviewer	2020

HotStorage '20 Program Committee	2020
SOSP '19 Artifact Evaluation Committee	2019
Eurosys '19 Shadow PC (**Best Reviewer**)	2019
ACM Transactions on Storage, Reviewer	2018
FAST '18, External Reviewer	2018
EuroSys '17, Contributor to PC Reviews	2017
OSDI '16, External Reviewer	2016
FAST '16, External Reviewer	2016

Presentations & Invited Talks

New Shared-Log Abstractions for Modern Applications
University of Texas, Austin — Oct '25

LazyLog: A New Log Abstraction for Low-Latency Applications
Current 2025 (hosted by Confluent) — Oct '25

LazyLog: A New Log Abstraction for Low-Latency Applications
Product Group at Confluent) — Sep '25

LazyLog: A New Log Abstraction for Low-Latency Applications
Data Streaming Summit 2025 (hosted by StreamNative) — Sep '25

New Shared-Log Abstractions for Modern Applications
Systems Distributed 2025 (hosted by TigerBeetle) — Jun '25

New Log Abstractions for Datacenter Applications
University of California, Berkeley — Oct '24
University of Pennsylvania — Nov '24

Co-designing Distributed Systems and Storage Stacks for Improved Reliability
University of Waterloo — Jan '22
Virginia Tech — Jan '22
Pennsylvania State University — Feb '22
University of Virginia — Feb '22
Purdue University — Feb '22
University of Utah — Feb '22
University of Toronto — Mar '22
University of Illinois at Urbana-Champaign — Mar '22
University of Washington — Mar '22
University of Michigan — Mar '22
Massachusetts Institute of Technology — Mar '22
University of North Carolina at Chapel Hill — Mar '22
University of Southern California — Mar '22
University of California, Santa Cruz — Mar '22
University of California, Irvine — Apr '22

Co-designing Distributed Systems and Storage Stacks
University of Waterloo (invited) — Oct '21

Reliable Distributed Storage: A Local-storage Perspective
Rutgers University (invited) — Aug '20

Reliable Distributed Storage: A Local-storage Perspective
VMware Research Group (postdoc interview talk) — Jun '20

Protocol-Aware Recovery for Consensus-Based Storage
Usenix ATC (invited conference talk) — Jul '19

Storage Systems at the Edge
NSF-VMWare ECDI Summit (invited) — Nov '18

Fault-Tolerance, Fast and Slow
Usenix OSDI (conference talk) — Oct '18

Protocol-Aware Recovery for Consensus-Based Storage
SNIA Storage Developer Conference (invited) — Sep '18

Resiliency to Storage Faults in Distributed Systems
Google Madison (invited) — May '18

Protocol-Aware Recovery for Consensus-Based Storage
Usenix FAST (conference talk) — Feb '18

Rethinking Consensus with Local Storage in Mind
SCI Labs Kickoff Meeting — May '17

Correlated Crash Vulnerabilities
Usenix OSDI (conference talk) — Oct '16

Correlated Crash Vulnerabilities
Microsoft Gray Systems Lab (invited) — Jun '16